

Homework

Please drive $\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{v^2} \frac{\partial^2(r\psi)}{\partial t^2}$ from the spherically symmetric differential wave equation: $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi(r,t)}{\partial r} \right) = \frac{1}{v^2} \frac{\partial^2 \psi(r,t)}{\partial t^2}$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi(r,t)}{\partial r} \right) = \frac{1}{v^2} \frac{\partial^2 \psi(r,t)}{\partial t^2}$$

both multiplied by r : $\frac{1}{r} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi(r,t)}{\partial r} \right) = \frac{1}{v^2} \cdot r \cdot \frac{\partial^2 \psi(r,t)}{\partial t^2}$

Left side: $\frac{1}{r} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi(r,t)}{\partial r} \right) = \frac{1}{r} \left[r^2 \frac{\partial^2 \psi}{\partial r^2} + 2r \frac{\partial \psi}{\partial r} \right] = r \frac{\partial^2 \psi}{\partial r^2} + 2 \frac{\partial \psi}{\partial r}$

Right side: $\because$ no correlations between r and $t \quad \therefore \frac{1}{v^2} \cdot r \cdot \frac{\partial^2 \psi(r,t)}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2(r\psi)}{\partial t^2}$

$\therefore \frac{\partial^2}{\partial r^2}(r\psi) = \frac{\partial}{\partial r} \left(r \frac{\partial \psi}{\partial r} + \psi \right) = r \frac{\partial^2 \psi}{\partial r^2} + \frac{\partial \psi}{\partial r} + \frac{\partial \psi}{\partial r} = r \frac{\partial^2 \psi}{\partial r^2} + 2 \frac{\partial \psi}{\partial r}$

$\therefore \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi(r,t)}{\partial r} \right) = \frac{1}{v^2} \frac{\partial^2 \psi(r,t)}{\partial t^2}$



$$\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{v^2} \frac{\partial^2(r\psi)}{\partial t^2}$$

Homework

□ 3.1, 3.2, 3.6 , 3.20

3.1 Consider the plane electromagnetic wave (in SI units) given by the expressions $E_x = 0$, $E_y = 2 \cos[2\pi \times 10^{14}(t - x/c) + \pi/2]$, and $E_z = 0$.

- (a) What are the frequency, wavelength, direction of motion, amplitude, initial phase angle, and polarization of the wave?
- (b) Write an expression for the magnetic flux density.

3.2 Write an expression for the $\vec{E}$ - and $\vec{B}$ -fields that constitute a plane harmonic wave traveling in the $+z$ -direction. The wave is linearly polarized with its plane of vibration at 45° to the yz -plane.

3.6 The electric field of an electromagnetic wave traveling in the positive x -direction is given by

$$\vec{E} = E_0 \hat{j} \sin \frac{\pi z}{z_0} \cos(kx - \omega t)$$

- (a) Describe the field verbally. (b) Determine an expression for k . (c) Find the phase speed of the wave.

3.20 A 1.0-mW laser has a beam diameter of 2 mm. Assuming the divergence of the beam to be negligible, compute its energy density in the vicinity of the laser.

3.1 Consider the plane electromagnetic wave (in SI units) given by the expressions $E_x = 0$, $E_y = 2 \cos[2\pi \times 10^{14}(t - x/c) + \pi/2]$, and $E_z = 0$.

- (a) What are the frequency, wavelength, direction of motion, amplitude, initial phase angle, and polarization of the wave?
- (b) Write an expression for the magnetic flux density.

3.1
$$E_y = 2 \cos [2\pi \times 10^{14} (t - x/c) + \pi/2]$$

$$E_y = A \cos [2\pi\nu(t - x/v) + \pi/2] \quad \text{from Eq. (2.26)}$$

- (a) $\nu = 10^{14}$ Hz, $v = c$, and $\lambda = c/\nu = 3 \times 10^8/10^{14} = 3 \times 10^{-6}$ m, moves in positive x -direction, $A = 2$ V/m, $\epsilon = \pi/2$ linearly polarized in y -direction.
- (b) $B_x = 0$, $B_y = 0$, $B_z = \frac{2}{c} \cos [2\pi \times 10^{14}(t - x/c) + \pi/2]$.

3.2 Write an expression for the $\vec{\mathbf{E}}$ - and $\vec{\mathbf{B}}$ -fields that constitute a plane harmonic wave traveling in the $+z$ -direction. The wave is linearly polarized with its plane of vibration at 45° to the yz -plane.

3.2 $E_z = 0$, $E_y = E_x = E_0 \sin(kz - \omega t)$ or cosine; $B_z = 0$, $B_y = -B_x = E_y/c$, or if you like,

$$\vec{\mathbf{E}} = \frac{E_0}{\sqrt{2}}(\hat{\mathbf{i}} + \hat{\mathbf{j}}) \sin(kz - \omega t), \quad \vec{\mathbf{B}} = \frac{E_0}{c\sqrt{2}}(\hat{\mathbf{j}} - \hat{\mathbf{i}}) \sin(kz - \omega t).$$

3.6 The electric field of an electromagnetic wave traveling in the positive x -direction is given by

$$\vec{E} = E_0 \hat{j} \sin \frac{\pi z}{z_0} \cos (kx - \omega t)$$

(a) Describe the field verbally. (b) Determine an expression for k . (c) Find the phase speed of the wave.

3.6 The field is linearly polarized in the y -direction and varies sinusoidally from zero at $z = 0$ to zero at $z = z_0$. Using the wave equation

$$\frac{\partial^2 E_y}{\partial x^2} + \frac{\partial^2 E_y}{\partial y^2} + \frac{\partial^2 E_y}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 E_y}{\partial t^2} = 0$$

$$\left[-k^2 - \frac{\pi^2}{z_0^2} + \frac{\omega^2}{c^2} \right] E_0 \sin \frac{\pi z}{z_0} \cos (kx - \omega t) = 0$$

and since this is true for all x , z , and t , each term must equal zero

$$\text{and so } k = \frac{\omega}{c} \sqrt{1 - \left(\frac{c\pi}{\omega z_0} \right)^2}$$

$$\text{Moreover, } v = \frac{\omega}{k} = \frac{c}{\sqrt{1 - \left(\frac{c\pi}{\omega z_0} \right)^2}}$$

3.20 A 1.0-mW laser has a beam diameter of 2 mm. Assuming the divergence of the beam to be negligible, compute its energy density in the vicinity of the laser.

$$\mathbf{3.20} \quad u = \frac{(\text{power})(t)}{(\text{volume})} = \frac{(10^{-3} \text{ W})(t)}{(\pi r^2)(ct)} = \frac{10^{-3} \text{ W}}{\pi(10^{-3})^2(3 \times 10^8)}$$

$$u = \frac{10^{-5}}{3\pi} \text{ J/m}^3 = 1.06 \times 10^{-6} \text{ J/m}^3$$

Homework

Problem 4.5

4.5 Imagine that we have a nonabsorbing glass plate of index n and thickness Δy , which stands between a source S and an observer P .

- (a) If the unobstructed wave (without the plate present) is $E_u = E_0 \exp i\omega(t - y/c)$, show that with the plate in place the observer sees a wave

$$E_p = E_0 \exp i\omega[t - (n - 1) \Delta y/c - y/c]$$

- (b) Show that if either $n \approx 1$ or Δy is very small, then

$$E_p = E_u + \frac{\omega(n - 1) \Delta y}{c} E_u e^{-i\pi/2}$$

The second term on the right may be envisioned as the field arising from the oscillators in the plate.

4.5 The phase angle is retarded by an amount $(n - 1) \Delta y 2\pi/\lambda$ or $(n - 1) \Delta y \omega/c$. Thus

$$E_p = E_0 \exp i\omega[t - (n - 1)\Delta y/c - y/c]$$

or $E_p = E_0 \exp [-i\omega(n - 1)\Delta y/c] \exp i\omega(t - y/c)$

if $n \approx 1$ or $\Delta y \ll 1$. Since $e^x \approx 1 + x$ for small x ,

$$\exp [-i\omega(n - 1)\Delta y/c] \approx 1 - i\omega(n - 1)\Delta y/c$$

and since $\exp (-i\pi/2) = -i$,

$$E_p = E_u + \frac{\omega(n - 1)\Delta y}{c} E_u e^{-i\pi/2}$$

Homework

Derive Fresnel equations

3.6.2 The Fresnel Equations

Case 1: $\vec{E}$ perpendicular to the plane-of-incidence

$$\vec{k} \times \vec{E} = v\vec{B}$$

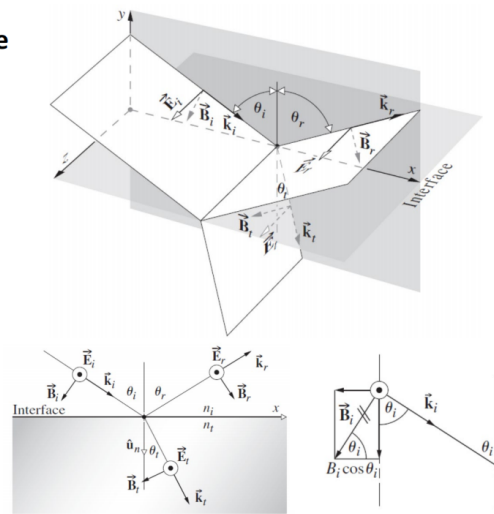
$$\vec{k} \cdot \vec{E} = 0$$

Continuity of the tangential components of the $\vec{E}$ -field at the boundary

$$E_{0i} + E_{0r} = E_{0t}$$

Continuity of the tangential components of the $\vec{B}/\mu$ at the boundary

$$-\frac{B_i}{\mu_i} \cos \theta_i + \frac{B_r}{\mu_i} \cos \theta_r = -\frac{B_t}{\mu_t} \cos \theta_t$$



Reflection and refraction of an incoming wave whose $\vec{E}$ -field is normal to the plane of incidence.

Fresnel equation (for any linear, isotropic, homogeneous dielectrics)

Amplitude reflection coefficient:

$$r_{\perp} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t}$$

Amplitude transmission coefficient:

$$t_{\perp} \equiv \left(\frac{E_{0t}}{E_{0i}} \right)_{\perp} = \frac{2n_i \cos \theta_i}{n_i \cos \theta_i + n_t \cos \theta_t}$$

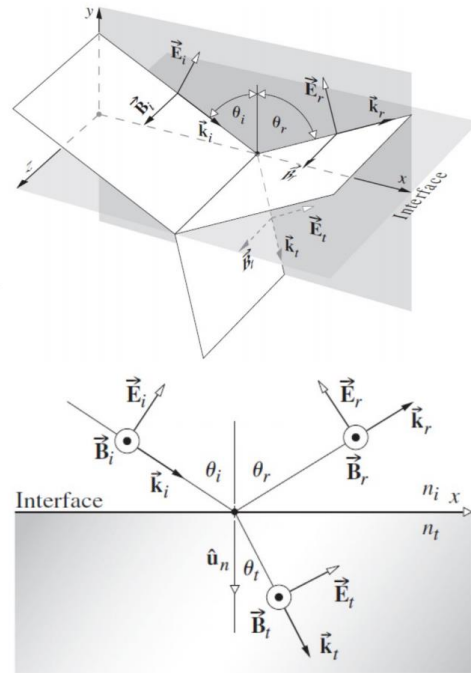
Case 2: $\vec{E}$ parallel to the plane-of-incidence

Continuity of the tangential components of the $\vec{E}$ -field at the boundary

$$E_{0i} \cos \theta_i - E_{0r} \cos \theta_r = E_{0t} \cos \theta_t$$

Continuity of the tangential components of the $\vec{B}/\mu$ at the boundary

$$\frac{B_i}{\mu_i} + \frac{B_r}{\mu_i} = \frac{B_t}{\mu_t}$$



Fresnel equation (for any linear, isotropic, homogeneous dielectrics)

Amplitude reflection coefficient:

$$r_{\parallel} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\parallel} = \frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_i \cos \theta_t + n_t \cos \theta_i}$$

$$r_{\perp} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t}$$

Amplitude transmission coefficient:

$$t_{\parallel} \equiv \left(\frac{E_{0t}}{E_{0i}} \right)_{\parallel} = \frac{2n_i \cos \theta_i}{n_i \cos \theta_t + n_t \cos \theta_i}$$

$$t_{\perp} \equiv \left(\frac{E_{0t}}{E_{0i}} \right)_{\perp} = \frac{2n_i \cos \theta_i}{n_i \cos \theta_i + n_t \cos \theta_t}$$

Further simplification using Snell's law

$$r_{\parallel} = + \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}$$

$$r_{\perp} = - \frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}$$

$$t_{\parallel} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t) \cos(\theta_i - \theta_t)}$$

$$t_{\perp} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t)}$$

Homework

▣ 4.14, 4.39, 4.50, 4.66, 4.70, 4.72, 4.76

4.14 A beam of 12-cm planar microwaves strikes the surface of a dielectric at 45° . If $n_{ti} = \frac{4}{3}$, compute (a) the wavelength in the transmitting medium, and (b) the angle θ_t .

$$\mathbf{4.14} \quad n_{ti} = \frac{n_t}{n_i} = \frac{c/v_t}{c/v_i} = \frac{v_i}{v_t} = \frac{\nu\lambda_i}{\nu\lambda_t} = \frac{\lambda_i}{\lambda_t}$$

therefore $\lambda_t = \lambda_i 3/4 = 9 \text{ cm}$

$$\sin \theta_i = n_{ti} \sin \theta_t$$

$$\sin^{-1} \left[\frac{3}{4}(0.707) \right] = \theta_t = 32^\circ$$

4.39 Derive Eqs. (4.42) through (4.45) for $r_{\perp}$, $r_{\parallel}$, $t_{\perp}$, and $t_{\parallel}$.

$$r_{\perp} = - \frac{\sin (\theta_i - \theta_t)}{\sin (\theta_i + \theta_t)} \quad (4.42)$$

$$r_{\parallel} = + \frac{\tan (\theta_i - \theta_t)}{\tan (\theta_i + \theta_t)} \quad (4.43)$$

$$t_{\perp} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin (\theta_i + \theta_t)} \quad (4.44)$$

$$t_{\parallel} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin (\theta_i + \theta_t) \cos (\theta_i - \theta_t)} \quad (4.45)$$

4.39 Starting with Eq. (4.34), divide top and bottom by n_i and replace n_{ti} with $\sin \theta_i / \sin \theta_t$ to get

$$r_{\perp} = \frac{\sin \theta_t \cos \theta_i - \sin \theta_i \cos \theta_t}{\sin \theta_t \cos \theta_i + \sin \theta_i \cos \theta_t}$$

which is equivalent to Eq. (4.42). Equation (4.44) follows in exactly the same way. To find $r_{\parallel}$ start the same way with Eq. (4.40) and get

$$r_{\parallel} = \frac{\sin \theta_i \cos \theta_t - \cos \theta_t \sin \theta_i}{\cos \theta_t \sin \theta_i + \sin \theta_i \cos \theta_t}$$

There are several routes that can be taken now: one is to rewrite $r_{\parallel}$ as

$$r_{\parallel} = \frac{(\sin \theta_i \cos \theta_t - \sin \theta_t \cos \theta_i)(\cos \theta_i \cos \theta_t - \sin \theta_i \sin \theta_t)}{(\sin \theta_i \cos \theta_t + \sin \theta_t \cos \theta_i)(\cos \theta_i \cos \theta_t + \sin \theta_i \sin \theta_t)}$$

$$\text{and so } r_{\parallel} = \frac{\sin (\theta_i - \theta_t) \cos (\theta_i + \theta_t)}{\sin (\theta_i + \theta_t) \cos (\theta_i - \theta_t)} = \frac{\tan (\theta_i - \theta_t)}{\tan (\theta_i + \theta_t)}$$

We can find $t_{\parallel}$, which has the same denominator, in a similar way.

4.50 Prove that

$$t_{\perp} + (-r_{\perp}) = 1 \quad [4.49]$$

for all θ_i , first from the boundary conditions and then from the Fresnel Equations.

4.50 $[E_{0r}]_{\perp} + [E_{0i}]_{\perp} = [E_{0t}]_{\perp}$; tangential field in incident medium equals that in transmitting medium,

$$[E_{0t}/E_{0i}]_{\perp} - [E_{0r}/E_{0i}]_{\perp} = 1, \quad t_{\perp} - r_{\perp} = 1$$

Alternatively, from Eqs. (4.42) and (4.44),

$$\frac{+\sin(\theta_i - \theta_t) + 2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t)} \stackrel{?}{=} 1$$

$$\frac{\sin \theta_i \cos \theta_t - \cos \theta_i \sin \theta_t + 2 \sin \theta_t \cos \theta_i}{\sin \theta_i \cos \theta_t + \cos \theta_i \sin \theta_t} = 1$$

4.66 Show that the polarization angles for internal and external reflection at a given interface are complementary, that is, $\theta_p + \theta'_p = 90^\circ$ (see Problem 4.64).

4.66

$$\tan \theta_p = n_t/n_i = n_2/n_1$$

$$\tan \theta'_p = n_1/n_2, \quad \tan \theta_p = 1/\tan \theta'_p$$

$$\frac{\sin \theta_p}{\cos \theta_p} = \frac{\cos \theta'_p}{\sin \theta'_p} \therefore \sin \theta_p \sin \theta'_p - \cos \theta_p \cos \theta'_p = 0$$

$$\cos (\theta_p + \theta'_p) = 0, \quad \theta_p + \theta'_p = 90^\circ$$

4.70 Show that

$$T_{\parallel} = \frac{\sin 2\theta_i \sin 2\theta_t}{\sin^2 (\theta_i + \theta_t) \cos^2 (\theta_i - \theta_t)} \quad (4.98)$$

and

$$T_{\perp} = \frac{\sin 2\theta_i \sin 2\theta_t}{\sin^2 (\theta_i + \theta_t)} \quad (4.99)$$

4.70 $T_{\perp} = \left(\frac{n_t \cos \theta_t}{n_i \cos \theta_i} \right) r_{\perp}^2$. From Eq. (4.44) and Snell's Law,

$$T_{\perp} = \left(\frac{\sin \theta_i \cos \theta_t}{\sin \theta_t \cos \theta_i} \right) \left(\frac{4 \sin^2 \theta_t \cos^2 \theta_i}{\sin^2 (\theta_i + \theta_t)} \right) = \frac{\sin 2\theta_i \sin 2\theta_t}{\sin^2 (\theta_i + \theta_t)}$$

Similarly for $T_{\parallel}$.

4.72 Suppose that we look at a source perpendicularly through a stack of N microscope slides. The source seen through even a dozen slides will be noticeably darker. Assuming negligible absorption, show that the total transmittance of the stack is given by

$$T_t = (1 - R)^{2N}$$

and evaluate T_t for three slides in air.

4.72 If Φ_i is the incident radiant flux or power and T is the transmittance across the first air–glass boundary, the transmitted flux is the $T\Phi_i$. From Eq. (4.68) at normal incidence the transmittance from glass to air is also T . Thus a flux $T\Phi_i T$ emerges from the first slide, and $\Phi_i T^{2N}$ from the last one. Since $T = 1 - R$, $T_t = (1 - R)^{2N}$ from Eq. (4.67).

$$R = (0.5/2.5)^2 = 4\%, \quad T = 96\%$$

$$T_t = (0.96)^6 \approx 78.3\%$$

4.76 Show that when $\theta_i > \theta_c$ at a dielectric interface, $r_{\parallel}$ and $r_{\perp}$ are complex and $r_{\perp}r_{\perp}^* = r_{\parallel}r_{\parallel}^* = 1$.

4.76 For $\theta_i > \theta_c$, Eq. (4.70) can be written

$$r_{\perp} = \frac{\cos \theta_i - i(\sin^2 \theta_i - n_{ii}^2)^{1/2}}{\cos \theta_i + i(\sin^2 \theta_i - n_{ii}^2)^{1/2}}$$

$$r_{\perp}r_{\perp}^* = \frac{\cos^2 \theta_i + \sin^2 \theta_i - n_{ii}^2}{\cos^2 \theta_i + \sin^2 \theta_i - n_{ii}^2} = 1$$

Similarly $r_{\parallel}r_{\parallel}^* = 1$.

Homework

Problems: 5.8, 5.13, 5.15, 5.24, 5.35, 5.37

5.8 Locate the image of an object placed 1.2 m from the vertex of a gypsy's crystal ball, which has a 20-cm diameter ($n = 1.5$). Make a sketch (of the rays, not the gypsy).

5.8 First surface: $\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$

$$\frac{1}{1.2} + \frac{1.5}{s_i} = \frac{0.5}{0.1}$$

$s_i = 0.36$ m (real image 0.36 m to the right of first vertex). Second surface $s_o = 0.20 - 0.36 = -0.16$ m (virtual object distance).

$$\frac{1.5}{-0.16} + \frac{1}{s_i} = \frac{-0.5}{-0.1}, \quad s_i = 0.069$$

Final image is real ($s_i > 0$), inverted ($M_T < 0$), and 6.9 cm to the right of the second vertex.

5.13 A biconcave lens ($n_l = 1.5$) has radii of 20 cm and 10 cm and an axial thickness of 5 cm. Describe the image of an object 1-inch tall placed 8 cm from the first vertex. Use the thin-lens equation to see how far off it is in determining the final-image location.

5.13 From Eq. (5.8), $1/8 + 1.5/s_i = 0.5/-20$. At first surface, $s_i = -10$ cm. Virtual image 10 cm to left of first vertex. At second surface, object is *real* 15 cm from second vertex.

$$1.5/15 + 1/s_i = -0.5/10, \quad s_i = -20/3 = -6.66 \text{ cm}$$

Virtual, to left of second vertex.

5.15 Prove that the minimum separation between conjugate *real* object and image points for a thin positive lens is $4f$.

5.15 $s_o + s_i = s_o s_i / f$ to minimize $s_o + s_i$,

$$\frac{d}{ds_o} (s_o + s_i) = 0 = 1 + \frac{ds_i}{ds_o}$$

$$\text{or } \frac{d}{ds_o} \left(\frac{s_o s_i}{f} \right) = \frac{s_i}{f} + \frac{s_o}{f} \frac{ds_i}{ds_o} = 0$$

$$\text{Thus } \frac{ds_i}{ds_o} = -1 \quad \text{and} \quad \frac{ds_i}{ds_o} = -\frac{s_i}{s_o}, \quad \therefore s_i = s_o$$

The separation would be maximum if either were ∞ , but both could not be. Hence, $s_i = s_o$ is the condition for a minima. From Gaussian equation, $s_o = s_i = 2f$.

5.24 The horse in Fig. 5.26 is 2.25 m tall, and it stands with its face 15.0 m from the plane of the thin lens whose focal length is 3.00 m.

- (a) Determine the location of the image of the equine nose.
- (b) Describe the image in detail—type, orientation, and magnification.
- (c) How tall is the image?
- (d) If the horse's tail is 17.5 m from the lens, how long, nose-to-tail, is the image of the beast?

5.24

- (a) From the Gaussian lens equation

$$\frac{1}{15.0 \text{ m}} + \frac{1}{s_i} = \frac{1}{3.00 \text{ m}}$$

and $s_i = +3.75 \text{ m}$.

- (b) Computing the magnification, we obtain

$$M_T = -\frac{s_i}{s_o} = -\frac{3.75 \text{ m}}{15.0 \text{ m}} = -0.25$$

Because the image distance is positive, the image is *real*. Because the magnification is negative, the image is *inverted*, and because the absolute value of the magnification is less than one, the image is *minified*.

- (c) From the definition of magnification, it follows that

$$y_i = M_T y_o = (-0.25)(2.25 \text{ m}) = -0.563 \text{ m}$$

where the minus sign reflects the fact that the image is inverted.

- (d) Again from the Gaussian equation

$$\frac{1}{17.5 \text{ m}} + \frac{1}{s_i} = \frac{1}{3.00 \text{ m}}$$

and $s_i = +3.62 \text{ m}$. The entire equine image is only 0.13 m long.

5.35 Verify Eq. (5.34), which gives M_T for a combination of two thin lenses.

$$\mathbf{5.35} \quad M_{T_1} = -s_{i1}/s_{o1} = -f_1/(s_{o1} - f_1)$$

$$M_{T_2} = -s_{i2}/s_{o2} = -s_{i2}/(d - s_{i1})$$

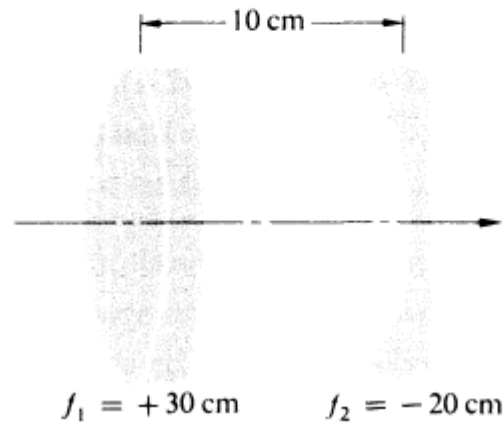
$$M_T = f_1 s_{i2}/(s_{o1} - f_1)(d - s_{i1})$$

From Eq. (5.30), on substituting for s_{i1} , we have

$$M_T = \frac{f_1 s_{i2}}{(s_{o1} - f_1)d - s_{o1}f_1}$$

5.37 Compute the image location and magnification of an object 30 cm from the front doublet of the thin-lens combination in Fig. P.5.37. Do the calculation by finding the effect of each lens separately. Make a sketch of appropriate rays.

Figure P.5.37

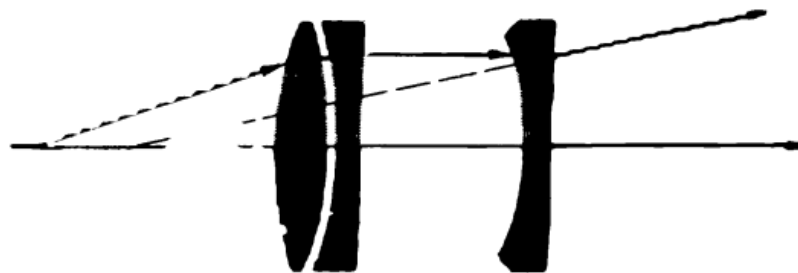


5.37 First lens $1/s_{i1} = 1/30 - 1/30 = 0$, $s_{i1} = \infty$. Second lens $1/s_{i2} = 1/(-20) - 1/(-\infty)$, the object for the second lens is to the right at ∞ , that is, $s_{o2} = -\infty$. $s_{i2} = -20$ cm, virtual, 10 cm to the left of first lens.

$$M_T = (-\infty/30) (+20/-\infty) = \frac{2}{3}$$

or from Eq. (5.34)

$$M_T = \frac{30(-20)}{10(30 - 30) - 30(30)} = \frac{2}{3}$$

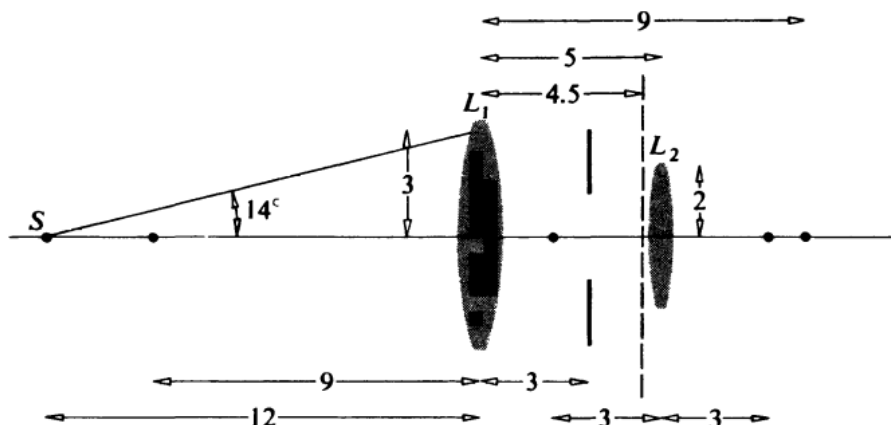


Homework

□ 5.44, 5.46, 5.47, 5.52, 5.55, 5.60

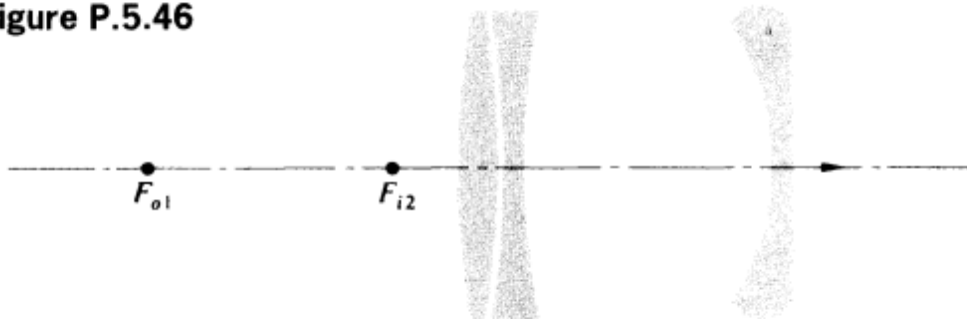
5.44 Consider the case of two positive thin lenses, L_1 and L_2 , separated by 5 cm. Their diameters are 6 and 4 cm, respectively, and their focal lengths are $f_1 = 9$ cm and $f_2 = 3$ cm. If a diaphragm with a hole 1 cm in diameter is located between them, 2 cm from L_2 , find (a) the aperture stop and (b) the locations and sizes of the pupils for an axial point, S , 12 cm in front of (to the left of) L_1 .

5.44 The angle subtended by L_1 at S is $\tan^{-1} 3/12 = 14^\circ$. To find the image of the diaphragm in L_1 we use Eq. (5.23): $x_o x_i = f^2$, $(-6)(x_i) = 81$, $x_i = -13.5$ cm, so that the image is 4.5 cm behind L_1 . The magnification is $-x_i/f = 13.5/9 = 1.5$, and thus the image (of the edge) of the hole is $(0.5)(1.5) = 0.75$ cm in radius. Hence the angle subtended at S is $\tan^{-1} 0.75/16.5 = 2.6^\circ$. The image of L_2 in L_1 is obtained from $(-4)(x_i) = 81$, $x_i = -20.2$ cm, in other words, the image is 11.2 cm to the right of L_1 . $M_T = 20.2/9 = 2.2$; hence the edge of L_2 is imaged 4.4 cm above the axis. Thus its subtended angle at S is $\tan^{-1} 4.4/(12 + 11.2)$ or 9.8° . Accordingly, the diaphragm is the A.S., and the entrance pupil (its image in L_1) has a diameter of 1.5 cm at 4.5 cm behind L_1 . The image of the diaphragm in L_2 is the exit pupil. Consequently $\frac{1}{2} + 1/s_i = \frac{1}{3}$ and $s_i = -6$, that is, 6 cm in front of L_2 . $M_T = \frac{6}{2} = 3$, so that the exit pupil diameter is 3 cm.

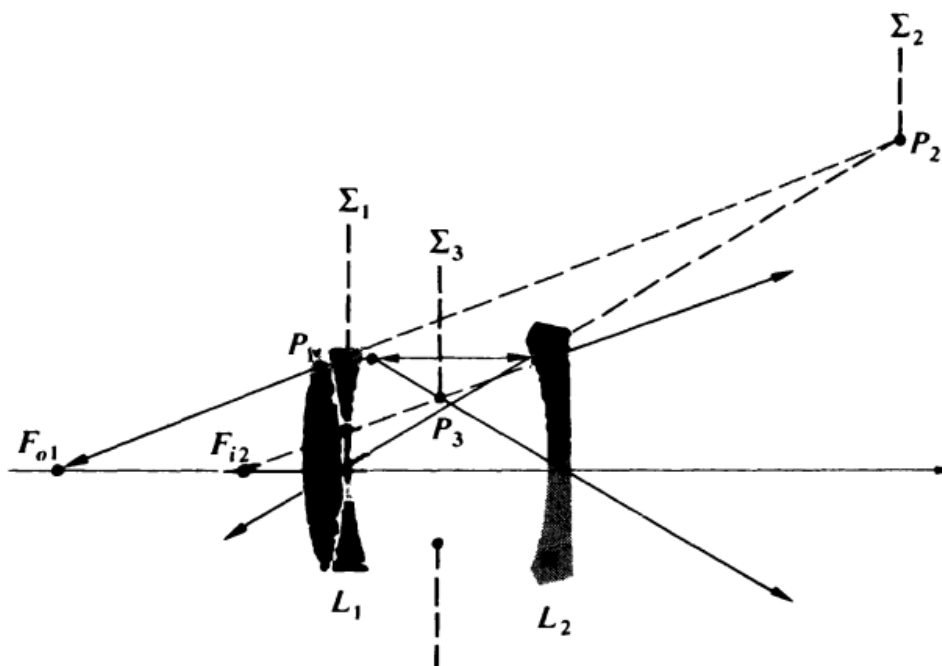


5.46 Make a sketch roughly locating the aperture stop and entrance and exit pupils for the lens in Fig. P.5.46, assuming the object point to be beyond (to the left of) F_{o1} .

Figure P.5.46

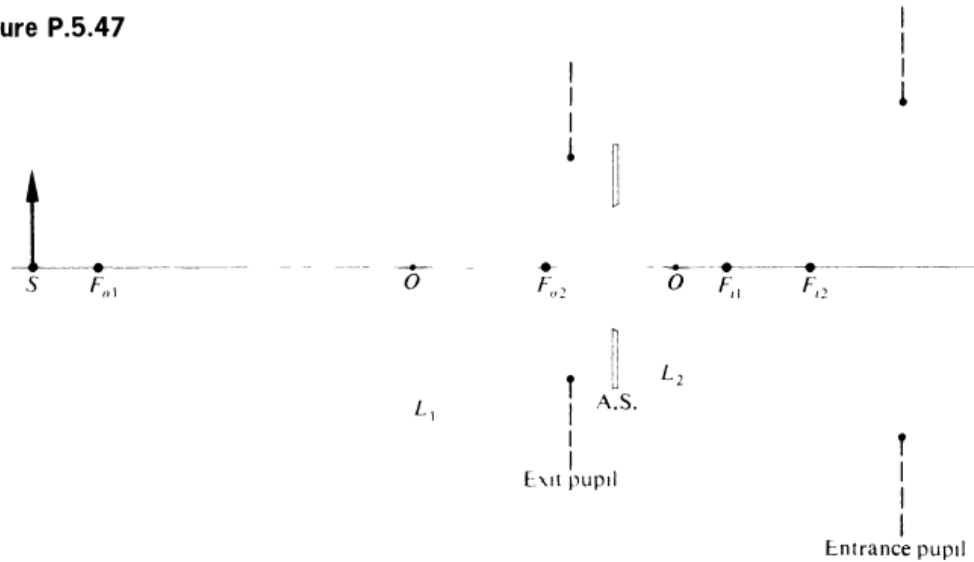


5.46 The A.S. is either the edge of L_1 or L_2 . Thus the entrance pupil is either marked by P_1 or P_2 . Beyond F_{o1} , P_1 subtends the smaller angle; thus Σ_1 locates the A.S. The image of the A.S. in the lenses to its right, L_2 , locates P_3 as the exit pupil.

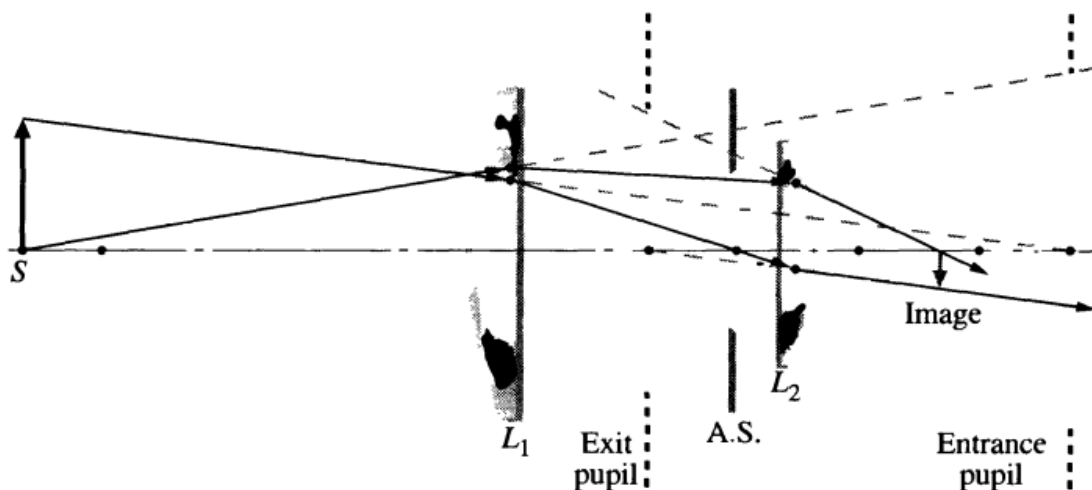


5.47 Figure P.5.47 shows a lens system, an object, and the appropriate pupils. Diagrammatically locate the image.

Figure P.5.47



5.47 Draw the chief ray from the tip to L_1 such that when extended it passes through the center of the entrance pupil. From there it goes through the center of the A.S., and then it bends at L_2 so as to extend through the center of the exit pupil. A marginal ray from S extends to the edge of the entrance pupil, bends at L_1 so it just misses the edge of A.S., and then bends at L_2 so as to pass by the edge of the exit pupil.



5.52 Locate the image of a paperclip 100 cm away from a convex spherical mirror having a radius of curvature of 80 cm.

5.52 From Eq. (5.49), $1/100 + 1/s_i = -2/80$, and so $s_i = -28.5$ cm. Virtual ($s_i < 0$), erect ($M_T > 0$), and minified. (Check with Table 5.5.)

5.55 The image of a red rose is formed by a concave spherical mirror on a screen 100 cm away. If the rose is 25 cm from the mirror, determine its radius of curvature.

5.55 Image on screen must be real $\therefore s_i$ is +

$$\frac{1}{25} + \frac{1}{100} = -\frac{2}{R}, \quad \frac{5}{100} = -\frac{2}{R}, \quad R = -40 \text{ cm}$$

5.60 An object is located at a distance s_o from a spherical mirror of radius R . Show that the resulting image will be magnified by an amount

$$M_T = \frac{R}{2s_o + R}$$

5.60 $M_T = y_i/y_o = -s_i/s_o$, using Eq. (5.50), $s_i = fs_o/(s_o - f)$, and since $f = -R/2$, $M_T = -f/(s_o - f) = -(-R/2)/(s_o + R/2) = R/(2s_o + R)$.

Homework

□ 5.71, 5.73

5.71 Determine the numerical aperture of a single clad optical fiber, given that the core has an index of 1.62 and the clad 1.52. When immersed in air, what is its maximum acceptance angle? What would happen to a ray incident at, say, 45° ?

5.71 From Eq. (5.64)

$$NA = (2.624 - 2.310)^{1/2} = 0.550$$

$$\theta_{\max} = \sin^{-1} 0.550 = 33^\circ 22'$$

Maximum acceptance angle is $2\theta_{\max} = 66^\circ 44'$. A ray at 45° would quickly leak out of the fiber; in other words, very little energy fails to escape, even at the first reflection.

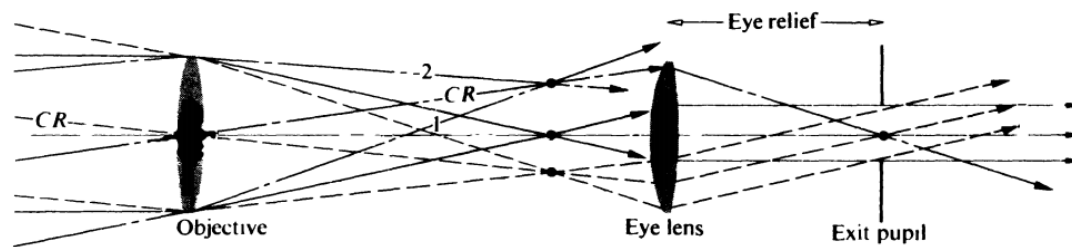
5.73 The number of modes in a stepped-index fiber is provided by the expression

$$N_m \cong \frac{1}{2}(\pi D \text{NA}/\lambda_0)^2$$

Given a fiber with a core diameter of $50 \mu\text{m}$ and $n_c = 1.482$ and $n_f = 1.500$, determine N_m when the fiber is illuminated by an LED emitting at a central wavelength of $0.85 \mu\text{m}$.

5.73 From Eq. (5.64) $\text{NA} = 0.232$ and $N_m = 9.2 \times 10^2$.

mula applied to the



Homework

□ 6.3, 6.6, 6.18, 6.19

6.3 Write an expression for the thickness d_l of a double-convex lens such that its focal length is infinite.

6.3 From Eq. (6.2), $1/f = 0$ when $-(1/R_1 - 1/R_2) = (n_l - 1)d/n_l R_1 R_2$. Thus $d = n_l(R_1 - R_2)/(n_l - 1)$.

6.6 Using Eq. (6.2), derive an expression for the focal length of a homogeneous transparent sphere of radius R . Locate its principal points.

6.6 $f = \frac{1}{2}nR/(n - 1); h_1 = +R, h_2 = -R$

6.18 Establish that Eqs. (6.36) and (6.37) are equivalent to Eqs. (6.3) and (6.4), respectively.

$$\begin{aligned}\mathbf{6.18} \quad h_1 &= n_{i1}(1 - a_{11})/-a_{12} = (\mathcal{D}_2 d_{21}/n_{t1})f \\ &= -(n_{t1} - 1)d_{21}f/R_2 n_{t1}\end{aligned}$$

from Eq. (5.64) where $n_{t1} = n_l$;

$$\begin{aligned}h_2 &= n_{i2}(a_{22} - 1)/-a_{12} \\ &= -(\mathcal{D}_1 d_{21}/n_{t1})f \text{ from Eq. (5.70)} \\ &= -(n_{i1} - 1)d_{21}f/R_1 n_{t1}\end{aligned}$$

6.19 Show that the planar surface of a concave-planar or convex-planar lens doesn't contribute to the system matrix.

6.19 $\mathcal{A} = \mathcal{R}_2 \mathcal{T}_{21} \mathcal{R}_1$, but for the planar surface

$$\mathcal{R}_2 = \begin{bmatrix} 1 & -\mathcal{D}_2 \\ 0 & 1 \end{bmatrix}$$

and $\mathcal{D}_2 = (n_{t1} - 1)/(-R_2)$ but $R_2 = \infty$

$$\mathcal{R}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

which is the unit matrix, hence $\mathcal{A} = \mathcal{T}_{21} \mathcal{R}_1$.

Homework

□6.26

6.26 Starting with the exact expression given by Eq. (5.5), show that Eq. (6.40) results, rather than Eq. (5.8), when the approximations for ℓ_o and ℓ_i are improved a bit.

6.26 From Eq. (5.6), let $\cos \varphi = 1 - \varphi^2/2$; then

$$\ell_o = [R^2 + (s_o + R)^2 - 2R(s_o + R) + R(s_o + R)\varphi^2]^{1/2}$$

$$\ell_o^{-1} = [s_o^2 + R(s_o + R)\varphi^2]^{-1/2}$$

$$\ell_i^{-1} = [s_i^2 - R(s_i - R)\varphi^2]^{-1/2}$$

where the first two terms of the binomial series are used,

$$\ell_o^{-1} \approx s_o^{-1} - (s_o + R)h^2/2s_o^3R \quad \text{where } \varphi \approx h/R,$$

$$\ell_i^{-1} \approx s_i^{-1} + (s_i - R)h^2/2s_i^3R$$

Substituting into Eq. (5.5) leads to Eq. (6.40).